

Joint Navigation and Online Aeromagnetic Compensation by Incremental Factor-Graph Smoothing

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Abstract— Airborne magnetic-anomaly navigation bounds inertial drift in GNSS-denied flight by matching a scalar magnetometer against an anomaly map. Its central obstacle is the aircraft’s own magnetic field, which must be compensated online when no calibration flight is available, the cold-start regime; on a cabin-mounted magnetometer the uncompensated field exceeds the full range of the map signal itself. We cast navigation and compensation as one maximum-a-posteriori problem on a factor graph, carrying the Tolles–Lawson coefficients as time-varying variables alongside the inertial error state, and solve it on board with an incremental fixed-lag smoother built on iSAM2, at a 300 s lag and 1 Hz states, 6.0 ms to 9.0 ms per update. The smoother produces two estimates and we report both throughout: a causal estimate with no look-ahead, which is the like-for-like comparison against recursive filters, and a committed estimate revised by the full lag, which is the output of a system permitted to lag real time. On the SGL 2020 flight data, across eight cold-start cases spanning four lines, two flights, two maps and two altitude regimes under one untuned configuration, the causal estimate alone beats an identically modelled online-TL EKF and a neural-network-augmented EKF on six of eight cases, and the committed estimate is the most accurate on all eight, at 10.5 m to 28.4 m where free-inertial drift is hundreds of metres. A lag sweep locates the design knee near 300 s and shows the causal estimate is indifferent to the lag, so look-ahead buys smoothing accuracy only; an ablation over every element of the configuration finds metre-level sensitivity and no delicate setting.

Index Terms— Magnetic-anomaly navigation, aeromagnetic compensation, Tolles–Lawson, factor-graph optimization, fixed-lag smoothing, GNSS-denied navigation.

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Nomenclature

$\mathbf{x}_t$	INS error state at epoch t (position, velocity, attitude, baro-inertial pair, IMU biases, FOGM), $\mathbf{x}_t \in \mathbb{R}^{18}$.
β_t	Online Tolles–Lawson compensation coefficients.
$\mathbf{x}_t$	Joint state $(\mathbf{x}_t, \beta_t)$.
$\Phi_t, \mathbf{Q}_t$	Pinson state-transition matrix and process covariance.
z_t, η_t	Scalar magnetometer measurement and its noise, variance R .
$h(\cdot)$	Anomaly-map interpolation (with IGRF core) as a function of position.
$\mathbf{g}_t$	Map gradient $\partial h / \partial \mathbf{p}$ at epoch t .
$\mathbf{A}_t$	Tolles–Lawson basis row from the three-axis fluxgate.
c_t	Aircraft magnetic interference, $c_t = \mathbf{A}_t^\top \beta_t$.
L_w, L_o	Sliding-window length and overlap (look-ahead).
ρ, δ	Robust kernel and its Huber threshold.
$\mathbf{G}, \Psi$	Stacked map-gradient and TL-regressor matrices over a window.
$\mathcal{N}(\cdot, \cdot)$	Gaussian density.
$\ \cdot\ _{\mathbf{M}}$	Mahalanobis norm, $\ \mathbf{v}\ _{\mathbf{M}}^2 = \mathbf{v}^\top \mathbf{M} \mathbf{v}$.

Boldface lowercase denotes vectors and boldface uppercase matrices.

I. Introduction

INERTIAL navigation systems (INS) are the backbone of airborne navigation, but their position error grows without bound as accelerometer and gyroscope errors are integrated over time, so an INS must be aided by an absolute position reference [1], [2]. The satellite-based reference of choice, GNSS, is unavailable or untrustworthy in the contested and signal-degraded environments that increasingly define aerospace and defense operations. This has renewed interest in passive, self-contained references that draw on natural geophysical signatures of the Earth. Among these, the crustal magnetic-anomaly field is attractive: it is globally available, temporally stable over the mission horizon, and observable with a lightweight scalar magnetometer [3], [4]. It is also difficult to spoof from a stand-off distance. Denial of a satellite signal exploits the far-field radio link, whose power falls off only as the inverse square of range, so a modest transmitter jams or spoofs a receiver from far away; a magnetic anomaly, by contrast, is a near-field source whose dipole field decays as the inverse cube of distance, so counterfeiting a credible anomaly at a moving aircraft would demand an impractically large or close field source [5]. The same map-matching idea underlies terrain-referenced and gravity-referenced navigation, where an onboard measurement is matched against a stored geophysical map to bound inertial drift [6], [7], [8]; magnetic-anomaly navigation is distinguished by the ubiquity and fine spatial structure of the crustal field and

by the maturity of airborne magnetic surveying. These properties make it a candidate absolute reference for long-endurance aircraft, uncrewed platforms, and munitions operating where GNSS cannot be trusted, provided the aircraft's own magnetic signature can be removed in flight.

Magnetic-anomaly navigation (MagNav) estimates aircraft position by matching the measured scalar field against a surveyed anomaly map, yielding an absolute fix that bounds inertial drift [4]. The idea has matured from feasibility study to flight demonstration: recursive Bayesian map matching, from the extended Kalman filter to grid and particle estimators, was characterized on flight data [3], [4] and flown on an F-16 [9], and recent multi-hour fixed-wing trials over thousands of kilometres report metre-to-decametre accuracy with quantum magnetometers [5]. As the estimators have matured, attention has moved to the inputs they rely on, and a consensus is forming that the pacing requirement for operational deployment is a sufficiently accurate, uncertainty-aware anomaly map [10]. Two obstacles, in fact, separate these demonstrations from routine use, and they are distinct: the reference map on the ground and the aircraft's own field in the air. The map is a survey-and-data problem, pursued elsewhere [10]; this paper addresses the second.

The platform's own permanent and induced magnetization and its eddy-current fields perturb the scalar measurement by hundreds of nanotesla, comparable to or larger than the anomaly signal being matched, so the aircraft field must be compensated before map matching can recover position. This aircraft-field problem is intrinsic to MagNav and independent of any particular airframe; it is most acute for magnetometers mounted inside the cabin, close to avionics and current-carrying structure, rather than on a wingtip stinger boom that holds the sensor away from the platform field.

Aeromagnetic compensation is classically handled by the Tolles–Lawson (TL) model [11], [12], [13], a regression of the interference onto sixteen attitude-dependent terms (permanent, induced, and eddy-current) formed from a three-axis fluxgate magnetometer. It faces two difficulties that are easy to conflate but are in fact orthogonal. The first is temporal: the coefficients are conventionally identified on a dedicated high-altitude calibration flight of prescribed maneuvers, yet many operational scenarios admit none, and the coefficients drift with temperature and configuration. In this cold-start regime they must be estimated online and jointly with navigation, from the same measurements that fix position, a coupling that makes both estimation and its analysis substantially harder than the calibrated case. The second is representational: however well its coefficients are fit, the linear TL basis cannot span interference from platform currents and other non-rigid, non-attitude sources, which contribute a large fraction of the field on cabin-mounted sensors [14], [15]. One difficulty is about identifying a known model class in time; the other is about that class being too small.

A public benchmark has made both concrete: the Signal Enhancement for Magnetic Navigation (SGL 2020) challenge dataset [14], with flight-grade INS, wingtip-stinger and cabin-mounted magnetometers, and high-resolution anomaly maps. Most work on it attacks the two difficulties offline: the TL coefficients by a batch calibration flight, and the representational gap by learned models fit offline on logged data, neural-network heading-error and denoising compensators [16], [15], physics-informed liquid-time-constant networks [17], and model-stability studies of the regression in gradiometry [18]. Offline learning, however, presupposes exactly what the operational scenario lacks: a corpus of logged flights from the same aircraft, a reference-grade compensated sensor flying alongside to supply training targets, the computation to train on that corpus, and a willingness to retrain whenever the platform configuration or its magnetic state drifts. A compensator fit this way is specific to one airframe in one equipment state, and its accuracy in flight is bounded by how well the training corpus transfers to the day's conditions. We therefore treat the offline family as context rather than competition: every estimator compared in this paper starts, as ours does, from no prior flight data. The exception is the online NN-in-EKF line of work originated by Gnadl [15], which carries the TL coefficients and the weights of a residual network as filter states learned in flight, and which has recently been extended with an error-state formulation and a natural-gradient interpretation [19]; this lineage holds, to our knowledge, the strongest reported cold-start results on the benchmark. This paper takes on the temporal difficulty fully: online, joint estimation of a time-varying compensation. The representational gap it does not model term-by-term: instead of a learned remainder, the estimator absorbs what the TL basis misses structurally, through the time-varying coefficients, a colored disturbance state, and a robust kernel, and Section V verifies against the recorded platform currents how much of the remainder this accounts for.

Recursive Bayesian filtering is the standard machinery for map-matching navigation, from the linear Kalman filter [20] to nonlinear variants. Geophysical map matching has a long lineage: terrain-contour matching for cruise missiles [6] and nonlinear Kalman filtering for terrain-aided navigation [7] established the paradigm, and grid- and particle-based Bayesian estimators later became standard where the measurement-to-position map is strongly nonlinear or multimodal [21], [22], [23], [8]. For MagNav specifically, Canciani and Raquet combined a calibrated scalar magnetometer with an extended Kalman filter (EKF) and characterized the achievable accuracy on flight data [3], [4], later demonstrating airborne magnetic navigation with online calibration [9]. Beyond the EKF, unscented filtering [24], invariant formulations that respect the state manifold [25], and mixture/particle representations [26] improve consistency under nonlinearity. However, these estimators are causal: they commit each state online with no access to future measurements, and

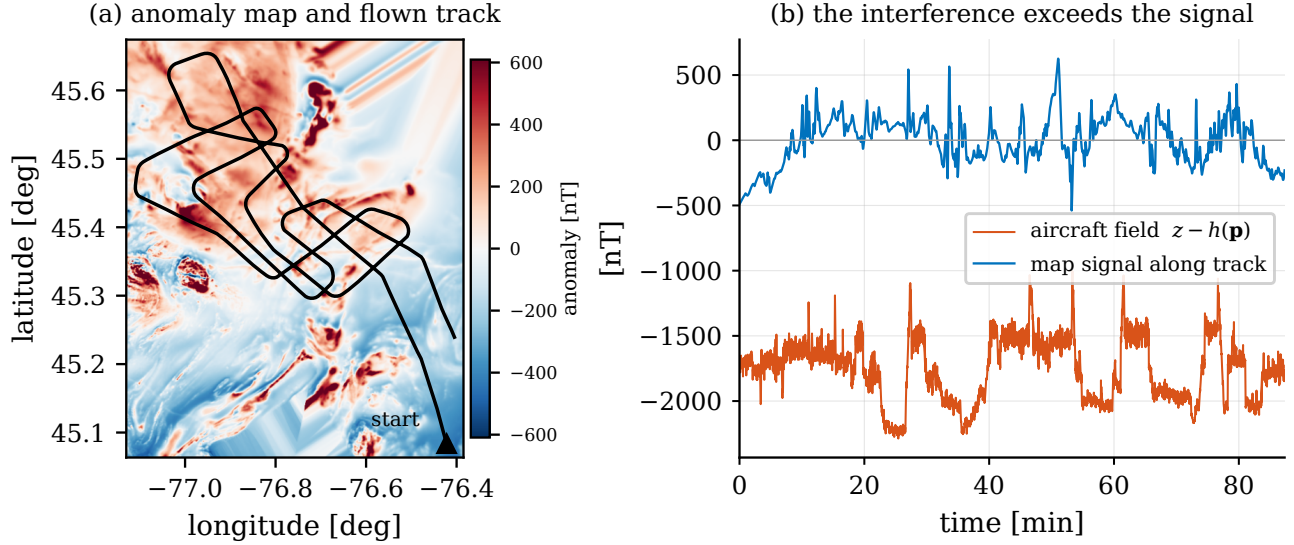


Fig. 1. The cold-start problem on line 1007.06. (a) Anomaly map and the flown track. (b) Measured scalar minus the map value along the track (the uncompensated aircraft field, Mag 4) against the map signal itself: the interference exceeds the signal’s full range.

the position-centric ones carry no compensation state of their own. When a large uncompensated cold-start residual enters, a causal filter has no mechanism to attribute the mismatch to the aircraft field rather than to position, and, unable to revise a committed state, the estimate can diverge.

A second line of work makes the compensation itself adaptive. Classical Tolles–Lawson (TL) identification is a batch regression on a calibration flight [11], [12], [13]; to remove that requirement the TL regression can be embedded in the estimator and its coefficients tracked online as additional states. Nonlinear extensions replace or augment the linear basis with learned components: neural-network compensators date back three decades [16], model-stability and robustness of the regression have been studied in gradiometry [18], and machine-learning-enhanced calibration has recently been applied to the SGL benchmark [15]. Most relevant here, a residual neural network (NN) has been added on top of an online TL model, carried inside an EKF and trained online, to capture interference the linear basis misses, introduced with Gnadt’s online models [15] and recently extended in [19]; this reaches good cold-start accuracy on the SGL 2020 cabin magnetometers and holds, to our knowledge, the strongest reported cold-start results on that benchmark. Nevertheless, the network requires a carefully engineered online-training design to remain stable, adds opaque parameters whose behavior is hard to certify, and, being embedded in an EKF, inherits the same causal, commit-online limitation as the filters above.

A third body of work, largely from robotics, replaces filtering with factor-graph optimization: the estimation problem is a graph of variables and probabilistic constraints [27] whose maximum-a-posteriori (MAP) solution is found by sparse nonlinear least squares [28], [29], the same structure underlying bundle adjustment and graph SLAM [30], [31], [32]. Incremental and square-

root smoothers (square-root SAM [33], iSAM [34], iSAM2 [35], and exactly-sparse delayed-state filters [36]) make the solution efficient enough for online use, and fixed-lag sliding-window optimization [37], [38] bounds its cost. On-manifold inertial preintegration [39], factor-graph inertial fusion [40], and modern visual–inertial estimators [41] have brought the approach firmly into navigation. Despite these advances, factor-graph estimation in navigation has been developed almost entirely for visual–inertial odometry and simultaneous localization and mapping, where the estimated quantities are poses and landmarks. The factor graph has very recently reached aeromagnetic compensation itself: Lathrop et al. [42] calibrate a magnetometer by fusing vector and scalar magnetometers with inertial measurements in a factor graph, absorbing large permanent moments and a time-varying external field. That work compensates the sensor but does not close the loop to navigation. The joint problem, estimating a time-varying compensation together with aircraft position from map matching, has not, to our knowledge, been posed in a published factor-graph formulation, and the observability of doing so has not been characterized.

Taken together, the three lines leave a specific gap. The cold-start MagNav problem is fundamentally a joint estimation of position and a time-varying compensation model from a single scalar stream, in which the two unknowns are confounded through the very measurement that must resolve them. Causal filters (cluster A) must commit each state before the maneuvers that would disambiguate the compensation have occurred, and so diverge when the initial residual is large; learned online compensation (cluster B) repairs that residual but keeps the causal, commit-online architecture and adds parameters that are difficult to certify; and factor-graph smoothing (cluster C), which can defer commitment and estimate structured nuisance variables jointly, has not been

applied to this problem, nor has anyone characterized when the joint problem is even observable. What is missing is an estimator that (i) defers commitment so that late-arriving calibration information corrects early navigation states, (ii) represents the compensation as first-class, time-varying variables rather than a black box, and (iii) comes with an explicit condition for when position and compensation can be separated at all. This paper supplies all three.

In this work we cast cold-start magnetic-anomaly navigation as a factor graph and, within it, carry the online Tolles–Lawson coefficients as time-varying variables, so that navigation and compensation are solved jointly as one maximum-a-posteriori (MAP) problem (Fig. 1). The graph is solved on board by an incremental fixed-lag smoother built on iSAM2 [35]: states are placed at 1 Hz, variables older than a 300s lag are marginalized, and each update touches only the part of the factorization the new measurement disturbs. Every state inside the lag is relinearized and revised as later measurements arrive, so late calibration information corrects early navigation states, the mechanism a causal filter lacks; a causal filter commits each state once and cannot revise it. We are explicit about scope: for a linear–Gaussian model the fixed-lag MAP estimate coincides with classical fixed-lag smoothing [43], so smoothing per se is not the contribution; the joint estimation of a time-varying compensation, delivered incrementally in real time, is. The paper makes the following contributions:

- 1) **Joint navigation and compensation on a factor graph.** We carry the online Tolles–Lawson coefficients as time-varying variables of a factor graph so that navigation and compensation are estimated jointly from the same scalar measurements. To the authors’ knowledge this is the first factor-graph formulation that jointly estimates time-varying Tolles–Lawson compensation and navigation states at a calibration cold start. Relinearizing the coefficients as the graph is re-solved makes the compensation adapt along the flight, the role a learned online model plays in the competing filter, while the graph keeps every quantity an explicit, physically interpretable state.
- 2) **A real-time incremental realization.** The graph is solved by an incremental fixed-lag smoother whose per-epoch cost is set by the lag, not the flight: 6.0 ms to 9.0 ms per state at a 300s lag and 1 Hz states, two orders of magnitude within the state cadence, in an unoptimized implementation. The lag is the single design variable, and we measure what it buys: the committed estimate improves steeply up to a knee near 300 s, about 20 km of track, while the causal estimate is indifferent to it.
- 3) **A causal/committed evaluation protocol.** A fixed-lag smoother produces two estimates, a causal one at the newest state and a committed

one revised by the full lag, and comparisons in the literature rarely distinguish them. We report both throughout, so the comparison against causal filters is like for like and the price of latency is explicit: the causal output alone beats the recursive baselines on most cases, and the committed output roughly halves its error at the cost of lagging real time by the lag.

- 4) **Flight-data validation.** On uncompensated cabin magnetometers across eight cold-start cases spanning four lines, two flights, two maps and two altitude regimes, under one untuned configuration, the committed estimate is the most accurate of the compared estimators on all eight cases and the causal estimate on six. Run rather than merely cited, the online EKF+TL+NN lineage of [15], [19] is kept bounded by its network where a plain online-TL EKF diverges, yet the smoother beats it with no learned component; an ablation over every element of the configuration finds metre-level sensitivity and no delicate setting.

The remainder of this paper is organized as follows. Section II states the problem in its classical batch form and formalizes the cold start. Section III sets out factor graph optimization from first principles. Section IV develops the joint system model, its factor graph, and the incremental fixed-lag solution. Section V describes the protocol and baselines and reports the experiments. Section VI concludes.

II. Problem Formulation

This section states what magnetic-anomaly navigation asks for in its classical batch form, shows why the aircraft’s own field prevents that form from being answered directly, and states the problem solved in the rest of the paper.

A. Batch map matching

Database-referenced navigation determines position by comparing a measured profile against a stored map. Let $\mathbf{p}_{0:k}$ be the horizontal positions of the aircraft over $k+1$ epochs, $z_{0:k}$ the scalar readings, and $h(\cdot)$ the map lookup returning the field predicted at a position. The classical batch estimate minimizes a profile-matching criterion, in mean absolute or mean squared form,

$$J_{\text{MAD}} = \frac{1}{k} \sum_{i=0}^k |z_i - h(\mathbf{p}_i)|, \quad J_{\text{MSD}} = \frac{1}{k} \sum_{i=0}^k (z_i - h(\mathbf{p}_i))^2, \quad (1)$$

subject to the inertial navigation system supplying the increments between consecutive positions,

$$\mathbf{p}_i - \mathbf{p}_{i-1} = \Delta \mathbf{p}_i^{\text{ins}}, \quad i = 1, \dots, k, \quad (2)$$

so that the search is over a rigid profile translated by the single unknown initial state,

$$\hat{\mathbf{p}}_{0:k} = \arg \min_{\mathbf{p}_{0:k}} J. \quad (3)$$

Terrain-referenced navigation is solved in this form routinely [44], [45], and the same form underlies the magnetic case [3], [4].

We state (1)–(3) as the reference formulation of database-referenced navigation rather than as a competing method: airborne magnetic-anomaly navigation is in practice run recursively, with a Kalman filter or a particle filter aiding the inertial solution [4], [15], and those are the baselines compared against in Section V. The batch form is worth writing down because it makes two things explicit that the recursive implementations leave implicit, and both matter here. It shows what the map contributes, namely the agreement between a measured profile and a stored one. And it treats the inertial increment as the hard constraint (2), which is the assumption the factor graph of Section III relaxes by admitting the increment as one more piece of uncertain information.

What (1) presumes is that the residual $z_i - h(\mathbf{p}_i)$ is small and unstructured once the position is right. For a scalar magnetometer mounted on an aircraft this is false, and by a wide margin. The measurement carries the aircraft's own magnetic field, generated by permanent magnetization, by currents induced in the airframe, and by eddy currents, all of which depend on the aircraft attitude in the Earth field and therefore change continuously along a line. On the five survey lines used in Section V the map value along the flown track varies with a standard deviation of 123 nT to 327 nT and a peak-to-peak swing of 586 nT to 2377 nT: that variation is the entire position signal. Against it, the uncompensated field at a cabin-mounted magnetometer is 1729 nT to 2081 nT root-mean-square over the first 300 s of flight. The interference is not a perturbation of the signal, it is larger than the signal's full range.

Compensation is therefore not a refinement but a precondition. The classical remedy is a calibration flight: a dedicated sequence of maneuvers at high altitude, where the anomaly field is weak, from which the interference model is fit once and then subtracted for the rest of the mission [11], [12], [13]. That remedy is unavailable in the situation this paper addresses. An aircraft that departs with an uncalibrated installation, or whose interference has changed since its last calibration, has no compensated measurement to feed into (1). Estimating position then requires estimating the compensation at the same time, from the same scalar stream that the map matching consumes, and the two are coupled through that single stream: an error in the compensation is indistinguishable, epoch by epoch, from an error in position. We refer to this situation as a cold start, and it is the setting of this paper.

B. Problem statement

The situation just described fixes what has to be estimated and what may be assumed. The aircraft carries a strapdown inertial navigation system whose error grows without bound, a three-axis fluxgate, and a scalar

magnetometer whose reading is corrupted by the aircraft field. A gridded anomaly map of the overflown region is available. No calibration flight has been made, so no compensation model is known at takeoff, and no satellite navigation is available at any point.

Problem 1 (Joint navigation and compensation at a cold start):

Given the scalar magnetometer measurements $z_{0:N}$, the fluxgate regressor rows $\mathbf{A}_{0:N}$, the anomaly map h , the inertial mechanization, and a prior $(\hat{\chi}_0, \mathbf{P}_0)$ that assumes no prior knowledge of the aircraft field, estimate the joint trajectory $\chi_{0:N}$ of inertial navigation errors $\mathbf{x}_k$ and time-varying compensation coefficients β_k , and do so with a per-epoch cost that does not grow with flight duration, so that the estimate is available on board.

Three properties of Problem 1 shape the rest of the paper. It is *joint*, because the compensation cannot be measured separately from the position it corrupts. It is *underdetermined at any single epoch*, because one scalar reading cannot separate a position error from a compensation error, so the two are told apart only over a span of epochs. And it is *online*, because an estimator that must see the whole flight before answering is of no use to the aircraft that flew it. The first two properties call for an estimator that carries the compensation as a state and revises past states as later data arrive; the third bounds how far into the past it may keep revising. Sections III and IV construct such an estimator.

III. Factor Graph Optimization

Problem 1 is stated as maximum a posteriori (MAP) inference and solved on a factor graph. This section sets out that machinery from the beginning, since the structure of the graph is what allows the compensation to be carried as a first-class unknown and the estimate to be produced in real time.

A. Variables, factors, and the graph

An estimation problem consists of quantities that are unknown and pieces of information that constrain them. In a factor graph the two are drawn as two kinds of node [46], [28]. A *variable node* holds an unknown, here the state χ_k at one epoch. A *factor node* holds one piece of information, and an edge joins a factor to every variable that piece of information involves. No edge joins two variables or two factors, so the graph is bipartite, and a factor is a function only of the variables it touches. Fig. 2 shows the pattern for a short trajectory.

Three kinds of factor appear in any navigation graph, and they exhaust the ones used in this paper. A *prior* factor is unary and states what is believed about a variable before any data: it fixes the otherwise free origin of the estimate. A *process* factor is binary and states how consecutive states are related through the system dynamics

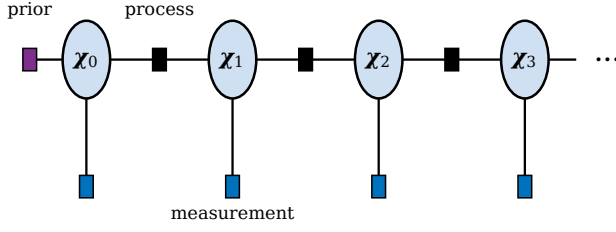


Fig. 2. Factor graph of a short trajectory. Circles are variable nodes, one per epoch; squares are factor nodes, each carrying one piece of information. The unary prior anchors the first state, binary process factors link consecutive states, and a unary measurement factor hangs from every state. A factor depends only on the variables it is joined to, which is what makes the problem sparse.

and the inertial increments. A *measurement* factor is unary and states what a sensor reading predicts about the state it was taken at. Writing $\mathbf{X} = \mathbf{x}_{0:N}$ for the collection of variables and $\mathbf{Z}$ for the data, the posterior factors along exactly these edges,

$$p(\mathbf{X} | \mathbf{Z}) \propto p(\mathbf{x}_0) \prod_{k=1}^N p(\mathbf{x}_k | \mathbf{x}_{k-1}) \prod_{k=0}^N p(\mathbf{z}_k | \mathbf{x}_k), \quad (4)$$

which is written compactly as a product of factors

$$J(\mathbf{X}; \mathbf{Z}) = \prod_i \phi_i(\mathbf{X}_i; \mathbf{Z}), \quad \hat{\mathbf{X}} = \arg \max_{\mathbf{X}} J(\mathbf{X}; \mathbf{Z}), \quad (5)$$

with $\mathbf{X}_i$ the variables adjacent to factor i . The graph is a picture of this factorization, and its sparsity is the problem's sparsity: each factor constrains one or two states, never the whole trajectory at once.

B. From factors to least squares

Each factor is modelled with a Gaussian kernel about its predicted value,

$$\phi_i(\mathbf{X}_i; \mathbf{Z}) \propto \exp\left(-\frac{1}{2}\|g_i(\mathbf{X}_i) - \mathbf{z}_i\|_{\mathbf{P}_i}^2\right), \quad (6)$$

where g_i is the prediction associated with the datum $\mathbf{z}_i$, $\mathbf{P}_i$ its covariance, and $\|\mathbf{e}\|_{\mathbf{P}} = \sqrt{\mathbf{e}^\top \mathbf{P}^{-1} \mathbf{e}}$ the Mahalanobis norm. Taking the negative logarithm of (5) turns the product into a sum and the maximization into a nonlinear least-squares problem,

$$\hat{\mathbf{X}} = \arg \min_{\mathbf{X}} \sum_i \|g_i(\mathbf{X}_i) - \mathbf{z}_i\|_{\mathbf{P}_i}^2. \quad (7)$$

The Mahalanobis weighting is what makes this sum meaningful. Inertial increments, magnetic readings in nanotesla, and compensation coefficients have different units and differ in magnitude by orders; they become comparable only after each residual is divided by its own uncertainty. The covariance attached to a factor is therefore not bookkeeping, it sets how much authority that piece of information carries against every other; Section D sets the one covariance a cold start is sensitive to.

C. Solving the graph

Because g_i is nonlinear in the states, (7) is solved iteratively. Linearizing every residual about the current estimate $\mathbf{X}^{(j)}$ and stacking gives

$$r(\mathbf{X}) = \begin{bmatrix} g_1(\mathbf{X}_1) - \mathbf{z}_1 \\ \vdots \\ g_{N_f}(\mathbf{X}_{N_f}) - \mathbf{z}_{N_f} \end{bmatrix}, \quad \mathcal{G} = \frac{\partial r}{\partial \mathbf{X}}, \quad (8)$$

after which the Levenberg–Marquardt step

$$\Delta \mathbf{X} = -(\mathcal{G}^\top \mathcal{G} + \lambda \mathbf{I})^{-1} \mathcal{G}^\top r(\mathbf{X}) \quad (9)$$

is applied and the residuals are relinearized. The damping λ interpolates between Gauss–Newton and gradient descent and makes the iteration robust where the problem is ill conditioned [33], which matters here because the map term is strongly nonlinear.

Two consequences follow, and they are the reason for preferring this formulation over a recursive filter. First, $\mathcal{G}$ inherits the sparsity of the graph: a chain of process factors with unary measurements gives a block-bidiagonal $\mathcal{G}$, so (9) costs time linear in the number of states rather than cubic. Second, every state is revised at every iteration. A filter commits each state once, at the moment it is reached, and can never revisit it; the graph relinearizes the whole set, so information that arrives late still corrects estimates made early. In the cold start of Section A this is exactly what is needed, because the compensation that explains the early measurements is only identified after the aircraft has manoeuvred.

D. Batch, fixed-lag, and incremental

The formulation says nothing about how much of the flight $\mathbf{X}$ contains, and that choice is the practical design variable. Taking $\mathbf{X}$ to be the whole line gives the batch estimate: it uses every measurement and is the most accurate, but it grows without bound and is available only after landing. Restricting $\mathbf{X}$ to the states inside a bounded interval of the present, and *marginalizing* those that fall outside, gives a fixed-lag smoother whose per-epoch cost does not grow with flight duration [40], [35]. Marginalization is not free: eliminating a variable replaces its factors by a single dense factor on its former neighbours, so what leaves the interval is summarized rather than discarded, and it can no longer be revised.

Rebuilding and re-solving that interval from scratch at every epoch would waste most of the work, since one new measurement changes little of a long chain. Incremental solvers instead update the factorization in place, touching only the part of it that the new data affect [35]. This is what makes a fixed-lag smoother run at sensor rate, and it is the solver this paper uses; Section C states the construction and Section E measures what the length of the interval costs and buys.

IV. Methodology

Section A left the cold start with two unknowns coupled through one scalar stream, and Section III supplied a formulation in which several unknowns of different kinds can be estimated together. This section puts them together: the system model first, then the factor graph it induces, then the incremental fixed-lag solution that produces an estimate on board.

A. System model

The estimated quantity at epoch k is the joint state

$$\mathbf{x}_k = \begin{bmatrix} \mathbf{x}_k \\ \boldsymbol{\beta}_k \end{bmatrix} \in \mathbb{R}^{18+m}, \quad (10)$$

stacking the inertial error state $\mathbf{x}_k$ and the compensation coefficients $\boldsymbol{\beta}_k$.

1. Inertial error state

We adopt the standard Pinson error model of a strap-down INS [1], [2], [47]. The error state

$$\mathbf{x} = [\delta \mathbf{p}^\top \delta \mathbf{v}^\top \boldsymbol{\psi}^\top h_a \hat{a} \mathbf{b}_a^\top \mathbf{b}_g^\top S]^\top \in \mathbb{R}^{18} \quad (11)$$

stacks position error $\delta \mathbf{p}$, velocity error $\delta \mathbf{v}$ and attitude error $\boldsymbol{\psi}$ (three components each), the baro-inertial altitude-aiding pair $(h_a, \hat{a})$ that damps the vertically unstable channel, accelerometer and gyroscope biases $\mathbf{b}_a, \mathbf{b}_g$, and a scalar first-order Gauss–Markov (FOGM) state S that absorbs residual magnetic disturbance and map error. Its discretized dynamics are

$$\mathbf{x}_k = \boldsymbol{\Phi}_{k-1} \mathbf{x}_{k-1} + \mathbf{w}_{k-1}, \quad \mathbf{w}_{k-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_{k-1}), \quad (12)$$

with $\boldsymbol{\Phi}_k$ the Pinson state-transition matrix (Appendix B) evaluated on the INS-indicated trajectory, and $\mathbf{Q}_k$ the process covariance. Unlike the hard constraint (2) of the batch form, the inertial increment enters as a factor with a covariance, so the estimator may disagree with the INS in proportion to how much that increment is trusted.

2. Compensation state

Classical aeromagnetic compensation writes the aircraft field as a linear combination of attitude-dependent basis terms [11], [12], [13],

$$c_k = \mathbf{A}_k^\top \boldsymbol{\beta}_k. \quad (13)$$

With direction cosines $\hat{\mathbf{b}}_k = [u \ v \ w]^\top$ of the Earth field in body axes, measured by a three-axis fluxgate, and total field magnitude B_k , the permanent (3), induced (6) and eddy-current (9) groups are

$$\mathbf{A}_k = \left[\underbrace{[u, v, w]}_{\text{perm.}}, \underbrace{[B_k u^2, \dots, B_k w^2]}_{\text{induced}}, \underbrace{[B_k u \dot{u}, \dots, B_k w \dot{w}]}_{\text{eddy}} \right]^\top, \quad (14)$$

optionally augmented by a constant term, giving $m \leq 19$ coefficients. In the calibrated setting $\boldsymbol{\beta}$ is fit once on a calibration flight and frozen. At a cold start it is unknown at takeoff, and the installation state drifts slowly in flight, so we carry it as a variable with a random-walk model

$$\boldsymbol{\beta}_k = \boldsymbol{\beta}_{k-1} + \mathbf{w}_{k-1}^\beta, \quad \mathbf{w}_{k-1}^\beta \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}^\beta). \quad (15)$$

This is the same device by which [48], [49] carry slowly varying sensor biases in a terrain-referenced graph. The difference is dimensional: a radar-altimeter bias is one scalar, whereas (14) is an m -dimensional regression whose columns are driven by aircraft attitude and therefore excited only by manoeuvre.

3. Measurement

The scalar magnetometer reads

$$z_k = h(\mathbf{p}_k) + \mathbf{A}_k^\top \boldsymbol{\beta}_k + S_k + \eta_k, \quad \eta_k \sim \mathcal{N}(0, R), \quad (16)$$

where $h(\cdot)$ interpolates the crustal anomaly map with the core field restored, evaluated at the true position $\mathbf{p}_k = \bar{\mathbf{p}}_k + \delta \mathbf{p}_k$ formed from the INS-indicated position and the estimated position error. Linearizing the map term about the INS-indicated position,

$$z_k \approx h(\bar{\mathbf{p}}_k) + \mathbf{g}_k^\top \delta \mathbf{p}_k + \mathbf{A}_k^\top \boldsymbol{\beta}_k + S_k + \eta_k, \quad \mathbf{g}_k = \left. \frac{\partial h}{\partial \mathbf{p}} \right|_{\bar{\mathbf{p}}_k}, \quad (17)$$

which exposes the structure of the problem. The map gradient $\mathbf{g}_k$ is the only channel through which the scalar reading carries horizontal position: where the map is locally flat, $\mathbf{g}_k \approx \mathbf{0}$ and position is momentarily unobservable. Everywhere else, the two terms $\mathbf{g}_k^\top \delta \mathbf{p}_k$ and $\mathbf{A}_k^\top \boldsymbol{\beta}_k$ compete for the same scalar: any position error can be mimicked by a compensation error whose regressor contribution matches it at that epoch. Separating them is possible only over a span of epochs, because $\mathbf{g}_k$ follows the map under the flight path while $\mathbf{A}_k$ follows the aircraft attitude, and the two vary differently. This is the reason a smoother is used rather than a filter, and it is quantified in Appendix A.

B. Factor graph formulation

The models of Section A induce four factor types on the variables $\boldsymbol{\chi}_{0:N}$, drawn in Fig. 3. Following the convention of (6), each is written as a squared Mahalanobis residual.

The *prior* factor anchors the initial joint state,

$$\bar{\phi}^0(\boldsymbol{\chi}_0) = \|\boldsymbol{\chi}_0 - \hat{\boldsymbol{\chi}}_0\|_{\mathbf{P}_0}^2, \quad (18)$$

where $\mathbf{P}_0$ is block diagonal in the inertial and compensation parts, with blocks $\mathbf{P}_0^x$ and $\mathbf{P}_0^\beta$.

The *inertial* factor relates consecutive states through (12),

$$\bar{\phi}_k^{\text{ins}}(\boldsymbol{\chi}_{k-1}, \boldsymbol{\chi}_k) = \|\mathbf{x}_k - \boldsymbol{\Phi}_{k-1} \mathbf{x}_{k-1}\|_{\mathbf{Q}_{k-1}}^2. \quad (19)$$

The *compensation* factor imposes the random walk (15),

$$\bar{\phi}_k^\beta(\boldsymbol{\chi}_{k-1}, \boldsymbol{\chi}_k) = \|\boldsymbol{\beta}_k - \boldsymbol{\beta}_{k-1}\|_{\mathbf{Q}^\beta}^2, \quad (20)$$

so that the compensation is free to adapt along the flight at a rate set by $\mathbf{Q}^\beta$, and is not free to jump.

The *map* factor is the only one that couples the two chains,

$$\bar{\phi}_k^{\text{map}}(\boldsymbol{\chi}_k) = \rho\left(\|h(\bar{\mathbf{p}}_k + \delta \mathbf{p}_k) + \mathbf{A}_k^\top \boldsymbol{\beta}_k + S_k - z_k\|_R^2\right), \quad (21)$$

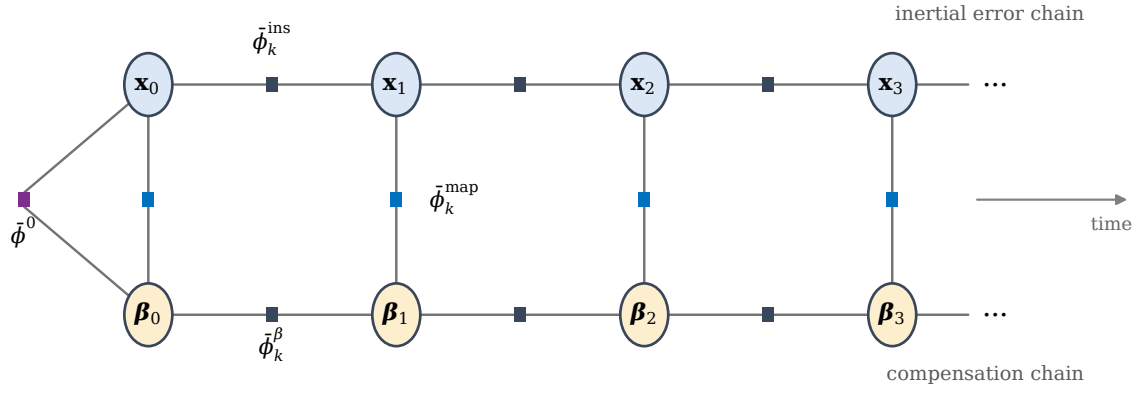


Fig. 3. Factor graph of the joint problem. The inertial error chain (top) and the compensation chain (bottom) evolve under their own process factors and are coupled at every epoch by the scalar map factor, which is the only place the two unknowns meet, and the reason they must be estimated together.

with ρ a robust kernel discussed below. Collecting the four types gives the objective

$$\tilde{J}(\mathbf{X}; \mathbf{Z}) = \bar{\phi}^0(\chi_0) + \sum_{k=1}^N \bar{\phi}_k^{\text{ins}} + \sum_{k=1}^N \bar{\phi}_k^{\beta} + \sum_{k=0}^N \bar{\phi}_k^{\text{map}}, \quad (22)$$

minimized as in (7). The graph has two parallel chains, the inertial error and the compensation, tied together at every epoch by a map factor, and it is this tie that makes the estimation joint rather than sequential.

Map error and transient interference produce heavy-tailed residuals that a purely quadratic cost over-weights. We take ρ to be the Huber kernel [50]

$$\rho(u) = \begin{cases} \frac{1}{2}u^2, & |u| \leq \delta, \\ \delta(|u| - \frac{1}{2}\delta), & |u| > \delta, \end{cases} \quad (23)$$

minimized by iteratively reweighted least squares, in which each iteration scales the map factor at epoch k by $w_k = \min(1, \delta/|u_k|)$ with $u_k = r_k/\sqrt{R}$, so that gross outliers are pulled toward an absolute-value penalty while inliers keep the efficient quadratic cost. We use $\delta = 1.345$, the standard choice for 95% Gaussian efficiency. Switchable constraints [51] and dynamic covariance scaling [52] fit the same factor interface and could replace (23) without touching the rest of the formulation.

C. Incremental fixed-lag solution

Minimizing (22) over the whole line is a batch problem, and by Section III it is available only after the flight. For onboard use we solve it at a fixed lag and incrementally.

1. Fixed lag

Let L be the lag, in seconds. At epoch k the graph retains the variables whose timestamps lie within L of the newest, and the variables older than that are marginalized: their information is compressed into a Gaussian factor on the oldest retained state, as described in Section III. Two

estimates are therefore produced at every epoch, and they are different products.

- The *causal* estimate is the value of the newest variable χ_k immediately after the update. It uses no measurement taken after k , so it is available with zero delay and is the estimate that a real-time navigator would read.
- The *committed* estimate is the value a variable holds when it leaves the lag, that is after it has been revised by every measurement in the following L seconds. It is the more accurate of the two and it is available only L seconds late.

Both are reported throughout Section V. Keeping them separate matters for comparison: a causal filter such as an EKF can only be compared against the causal estimate, whereas the committed estimate is the output of a system that is allowed to lag real time by L .

2. Incremental update

Re-solving the retained interval from scratch at every epoch would repeat almost all of the previous work, since one new measurement disturbs only a small part of a long chain. We instead update the factorization in place using the Bayes-tree formulation of iSAM2 [35]. At each epoch the inertial factor (19), the compensation factor (20) and the map factor (21) of that epoch are added; iSAM2 identifies the variables whose linearization the new factors disturb, re-eliminates only that part of the tree, and marginalizes what has left the lag. The per-epoch cost is therefore set by the number of retained states, L divided by the state interval, and not by the flight duration.

3. State interval

Placing one variable per magnetometer sample makes the retained set large for any useful lag. We therefore place variables at an interval Δ longer than the sampling period, compounding the transition and accumulating the process noise exactly across the K samples that a variable

TABLE I

Flight lines. FF = free flight, SV = survey; maps R = Renfrew, E = Eastern; INS = free-inertial drift [m].

Line	Purp.	Map	alt. [m]	dur. [min]	INS [m]
1003.02	SV	E	401	63.1	123.8
1003.08	SV	R	400	72.4	271.9
1007.02	FF	E	799	64.3	120.7
1007.06	FF	R	402	87.3	317.5

interval spans,

$$\Phi' = \Phi_{k+K-1} \cdots \Phi_k, \quad \mathbf{Q}' \leftarrow \Phi \mathbf{Q}' \Phi^\top + \mathbf{Q}, \quad (24)$$

so that the retained chain is an exact marginal of the full-rate chain in its process model. The map factors at the intermediate samples are not carried; the cost of that choice is measured in Section G.

4. Conditioning

One numerical point matters for reproduction. The position channel of $\mathbf{Q}_k$ is minute, of order 10^{-31} rad^2 in the horizontal states, because position error is driven only through velocity over a single sample; whitening the inertial factor by $\mathbf{Q}_k^{-1/2}$ then spans some thirty orders of magnitude. A Cholesky factorization of the information matrix squares this conditioning and fails outright, returning an indeterminate variable. Two changes remove the failure: a floor of $10^{-20} \mathbf{I}$ added to $\mathbf{Q}_k$, which injects under a millimetre of position uncertainty per step and is dynamically negligible, and a QR factorization of the square-root information matrix in place of Cholesky, which works on the square root and so tolerates the residual spread [33]. A larger floor, or a regularizing prior at every epoch, also suppresses the exception but pins the estimate onto the propagated INS and defeats the correction.

D. Prior selection

One element of (18) is set with the cold start in mind. The block $\mathbf{P}_0^\beta$ states what is believed about the compensation coefficients before any measurement, and at a cold start that belief must admit the field of an uncalibrated installation: the uncompensated cabin field is of order 2000 nT, and the coefficients that explain it are far from zero in the units of (14). For initial stability we therefore widen the compensation block relative to a calibrated installation and use $\sigma_\beta = 100$ throughout. The setting is not delicate: the ablation of Section F varies it by an order of magnitude in each direction, and every other element of the configuration, and finds metre-level effects.

V. Experiments and Results

A. Dataset, maps, and protocol

All experiments use the public SGL 2020 challenge dataset [14]: a Cessna 208B instrumented with a flight-

TABLE II

Estimator settings, common to every line and both magnetometers.

Group	Parameter	Value
Measurement	noise std. $\sqrt{R}$	12 nT
	robust kernel, Huber δ	1.345
FOGM S	(σ_S, τ)	3 nT, 180 s
Initial $\mathbf{P}_0^x$	pos. / alt. std.	0.1 / 1.0 m
	velocity std. (per axis)	1.0 m s^{-1}
Compensation	basis (App. C)	P+I+E+bias ($m = 19$)
	prior $\mathbf{P}_0^\beta = \mathbf{I} \sigma_\beta^2$	$\sigma_\beta = 100$
	walk $\mathbf{Q}^\beta$	$\mathbf{I}(1.0)^2 \Delta t$
Smoother	lag L	300 s
	state interval Δ	1 s ($K = 10$)
	solver	iSAM2, QR
	process-noise floor	$10^{-20} \mathbf{I}$
Evaluation	DRMS warm-up	600 s

grade INS, several onboard scalar magnetometers (a compensated tail-stinger sensor and uncompensated cabin sensors Mag 4 and Mag 5, all optically pumped cesium-vapor units), a three-axis fluxgate, and reference truth. The flights serve distinct purposes, and we choose lines accordingly (Table I). Flt1007 is the free-flight mission, in which the aircraft flies freely within the mapped areas, so its lines are the closest to a navigation task and are what prior work [14], [4], [19] reports; 1007.06 is our representative line for that reason. Flt1003 is a crustal-field survey flight, and its lines 1003.02 and 1003.08 are straighter than the free-flight legs. Flt1006 is a calibration-manoeuve flight and, with the map-building flights (Flt1004–1005), is not used: its single 14 min line leaves too little to score after the warm-up. Two honest points remain. None of these lines is a dedicated long, straight-and-level operational transit; the free-flight and survey legs carry attitude variation, which stresses the compensation but also aids its observability by exciting the regressor, so performance on a pure operational cruise is not established here. And the lines span two altitude regimes, four near the 400 m survey-drape altitude and 1007.02 near 800 m, where upward continuation attenuates the anomaly.

Map matching uses the Eastern and Renfrew high-resolution anomaly grids, upward-continued to flight altitude, with the IGRF core field restored so that measurement and map share an absolute scale. Cold start means the compensation coefficients are initialized at zero with no calibration flight; the navigation solution itself is accurately initialized (position standard deviation 0.1 m, Table II), so the term denotes a compensation cold start, not an unknown-position start.

The error metric is the horizontal distance root-mean-square (DRMS) of position error, counted after a 600 s warm-up. Following Section C, the proposed estimator is reported twice in every table: the *causal* value, read at the newest state with no look-ahead, and the *committed* value, read as states leave the 300 s lag. The causal value is the like-for-like comparison against the recursive baselines, which also use no future data; the committed value is

the output of a system permitted to lag real time by the lag length, and the look-ahead behind every reported number is stated in its table. One configuration (Table II) is used for every line and both magnetometers; nothing is tuned per case, and parameter sensitivity is quantified in the ablation of Section F. All numbers are produced by scripted runs under a fixed software environment and are available from the authors on request.

B. Baselines

Four references frame the comparison, all run by us on the same lines, the same maps, and the same inertial model.

Free-inertial INS. The unaided INS drift (Table I) is the floor every aided estimator must beat, and the common reference that makes DRMS values comparable across lines of different length and roughness.

Online-TL EKF. The direct recursive counterpart of the proposed estimator: the same Pinson error state augmented with the same 19 time-varying TL coefficients, updated by the same scalar measurement, in an extended Kalman filter [15]. It shares every model term with the factor graph, so the margin between the two isolates the estimator structure, recursive commitment against windowed re-linearization, from the modelling.

Online EKF+TL+NN. The strongest published causal method for this dataset augments the TL state with the weights of a small feed-forward network that absorbs what the linear basis cannot [15], [19]. We run the reference implementation of [15], tuned by us for a stable cold start, and do not claim to reproduce the extended filter of [19]. Cold start forces one substantive change: with no calibration flight to warm-start from, the network output normalization is initialized from the first five minutes of onboard interference; a naive port instead drives the network bias to $\sim 5 \times 10^4$ nT and the filter to NaN. We keep the uncompensated scalar out of the network inputs: a position-dependent input carries the map gradient, so the network absorbs map signal and position drifts while the residual stays small (the scalar-augmented variant of [19] escapes this only by freezing the network after its transient). The tuned filter uses the permanent TL group with an eight-unit hidden layer and reaches 48.8/18.3 m on 1007.06 Mag 4/5, consistent with published cold-start operating points [15], [19].

Marginalized particle filter (MPF+TL). The nonparametric reference, included because particle methods are the standard resort when the map term is strongly non-linear [8], [49]. Since the TL model is linear in its coefficients, β is conditionally linear-Gaussian and fits the Rao-Blackwellized structure exactly: our implementation carries position in 1000 particles and marginalizes the remaining states, including β , in the conditional Kalman block, with the fluxgate row entering the linear measurement Jacobian. Weights are computed in the log domain and position diversity is reinjected after resampling. Despite this, and under per-case tuning of measurement

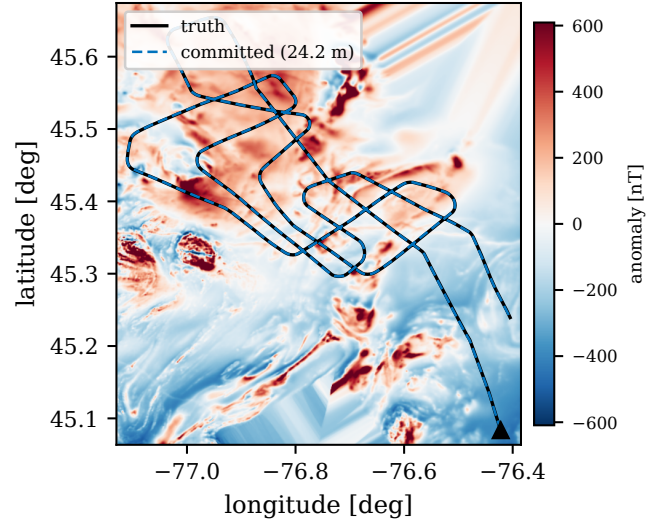


Fig. 4. Line 1007.06: committed estimate against truth on the anomaly map.

TABLE III

Line 1007.06 cold start: horizontal DRMS [m] after a 600 s warm-up. The causal output uses no look-ahead; the committed output lags by 300 s.

Estimator	Mag 4	Mag 5	look-ahead
EKF, online TL	46.7	17.8	none
EKF+TL+NN [15]	48.8	18.3	none
MPF+TL (best tuning)	1227.0	75.1	none
Proposed, causal	42.7	16.0	none
Proposed, committed	24.2	11.5	300 s

variance, resampling threshold and roughening that no other estimator in this paper receives, the filter is unstable at a cold start: its best per-case results range from 38.7 m on the cleanest Mag 5 leg to kilometres on Mag 4, with strong seed dependence. We report it in the text rather than in the comparison tables, and we do not claim it represents the best achievable particle filter. Attaching the neural network of the EKF+TL+NN to the particle filter does not rescue it and is sharply harmful on the well-conditioned sensor, degrading Mag 5 by one to two orders of magnitude across lines, the same direction seen for the NN-augmented EKF where the linear basis already suffices.

C. Representative line

Table III reports the longest line, 1007.06 (87 min), on both cabin magnetometers, with the free-inertial drift at 317.5 m. Read causally, with no look-ahead at all, the proposed estimator reaches 42.7 m on Mag 4 and 16.0 m on Mag 5, ahead of the online-TL EKF (46.7/17.8 m) and of the NN-augmented filter (48.8/18.3 m). The margin over the online-TL EKF is the purest number in this paper: the two estimators share the error model, the compensation model, the measurement model, and every noise

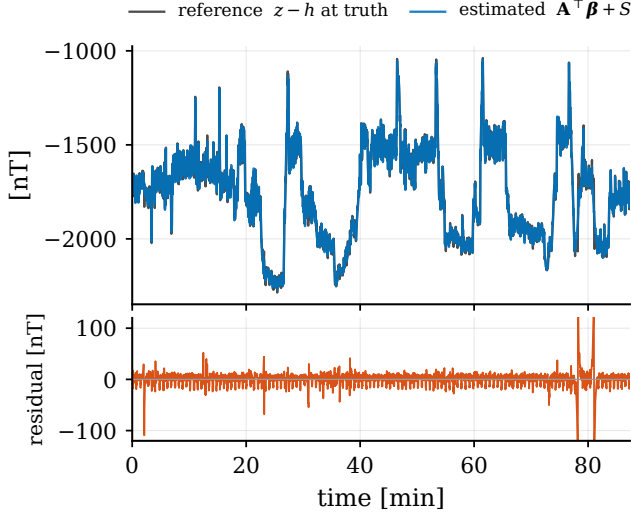


Fig. 5. What the compensation chain learned (Mag 4): the estimated $\mathbf{A}^\top \boldsymbol{\beta} + S$ overlays the interference reference; the residual is tens of nanotesla.

TABLE IV

Breadth: DRMS [m] after a 600 s warm-up. “div.” > 10 km, “err.” leaves the map; bold = best causal entry.

Line	Mag	EKF		MPF+		Proposed	
		online	TL+NN	TL	causal	300 s	
1007.06	4	46.7	48.8	1227.0	42.7	24.2	
1007.06	5	17.8	18.3	75.1	16.0	11.5	
1007.02	4	div.	130.0	1311.5	74.3	28.4	
1007.02	5	31.6	30.4	2247.3	33.1	15.8	
1003.02	4	div.	101.0	482.6	58.2	19.6	
1003.02	5	28.1	29.5	40.9	21.9	12.0	
1003.08	4	err.	46.6	2898.7	48.9	25.0	
1003.08	5	21.1	20.7	77.1	19.2	10.5	
best causal					6 / 8		
best overall						8 / 8	

setting, and differ only in that the filter commits each state once while the graph relinearizes and revises every state inside its lag. The margin over the EKF+TL+NN says something further: the network exists to absorb what the linear TL basis cannot, yet carrying the linear basis in a smoother beats carrying the linear basis plus a network in a filter, on both sensors, without any learned component. Allowed its 300 s lag, the committed estimate roughly halves the causal error, to 24.2/11.5 m. Fig. 7 shows the error history, Fig. 6 its north and east components against every baseline, Fig. 4 the recovered track on the map, and Fig. 5 the compensation the graph learned while recovering it.

D. Breadth

Table IV repeats the comparison on every counted line, spanning two flights, two maps, two altitude regimes, and both cabin magnetometers, with nothing re-tuned (Fig. 8). The committed estimate is the most accurate

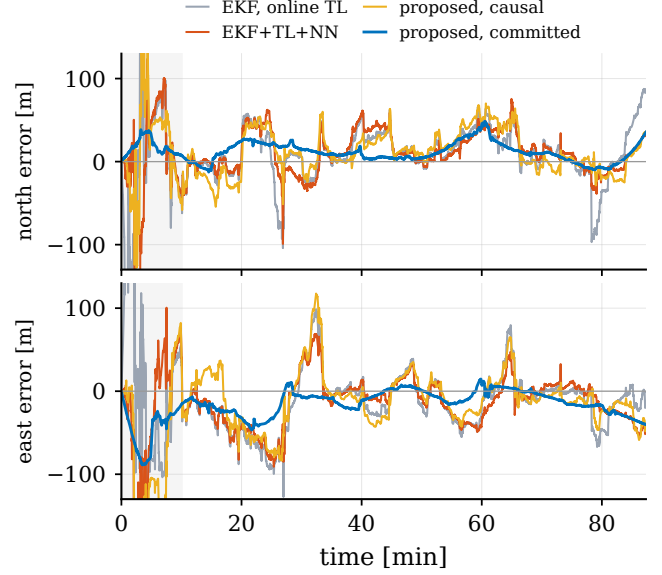


Fig. 6. North and east error histories of the compared methods on line 1007.06 Mag 4. The EKF+TL+NN trace is a fresh run of the seed-sensitive filter (40.0 m on this run, against the fixed reference run of Table III).

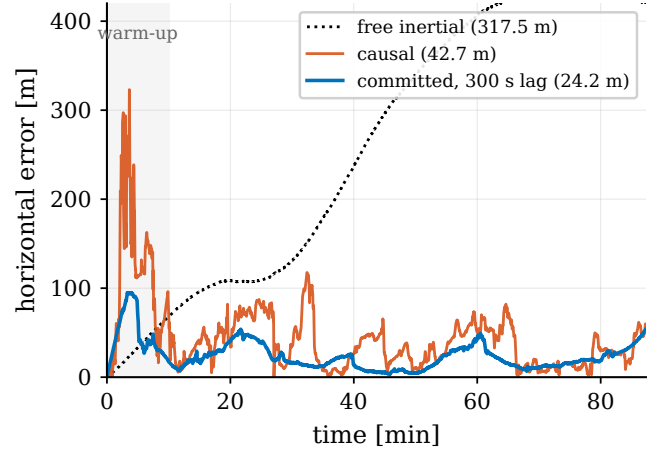


Fig. 7. Line 1007.06 Mag 4, cold start: horizontal error of the causal and the committed estimates against free-inertial drift. The committed trace is the causal trace revised by 300 s of look-ahead before each state is frozen.

entry on all eight counted cases, between 10.5 and 28.4 m, and between 1.5 and 5.2 times better than the strongest causal baseline on the same case. Read causally, the proposed estimator is the best causal method on six of the eight; it trails the EKF+TL+NN on 1007.02 Mag 5 by 2.7 m and on 1003.08 Mag 4 by 2.3 m, and we report both rather than claiming a sweep. Where the online-TL EKF fails outright, diverging on two Mag 4 lines and leaving the map on a third, both readings of the proposed estimator stay bounded; the mechanism is the one Section A anticipated, since the filter must explain the uncompensated field with whatever state is cheapest at

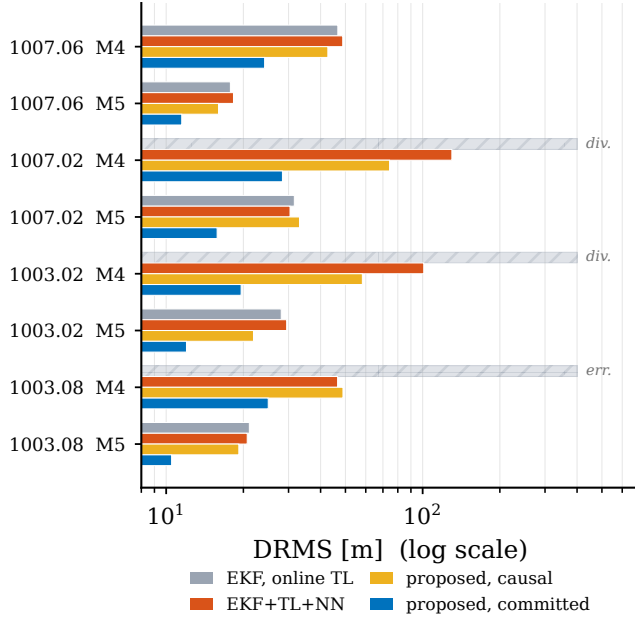


Fig. 8. Breadth comparison of Table IV on a log axis.

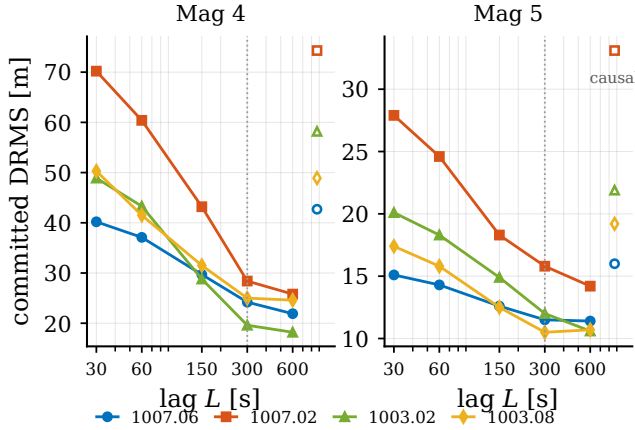


Fig. 9. Committed DRMS against lag (Table V); open markers show the lag-independent causal error.

each instant, while the smoother may revise its attribution for 300 s before committing.

E. Lag design

The lag is the estimator's single design variable, and Table V sweeps it from 30 to 600 s on the four counted lines (Fig. 9). Two regularities carry the design guidance. First, the committed error improves with lag on every case, steeply up to about 300 s and weakly after it: lengthening the lag from 300 to 600 s, a doubling of both latency and computation, buys at most 2.6 m and on one case slightly degrades, while the step from 30 to 300 s buys between 3.6 and 41.8 m. The knee reflects the physics: at survey speed a 300 s lag corresponds to roughly 20 km of track, enough for the map gradient and

TABLE V

Lag sweep: committed DRMS [m], causal at $L = 300$ s. The causal output varies by under a metre across the sweep.

Line	Mag	lag L [s], committed					causal
		30	60	150	300	600	
1007.06	4	40.2	37.1	29.7	24.2	21.9	42.7
1007.06	5	15.1	14.3	12.6	11.5	11.4	16.0
1007.02	4	70.2	60.4	43.2	28.4	25.8	74.3
1007.02	5	27.9	24.6	18.3	15.8	14.2	33.1
1003.02	4	48.9	43.3	28.8	19.6	18.2	58.2
1003.02	5	20.1	18.3	14.9	12.0	10.6	21.9
1003.08	4	50.3	41.5	31.5	25.0	24.6	48.9
1003.08	5	17.4	15.8	12.5	10.5	10.7	19.2

TABLE VI

Ablation on line 1007.06, one change per row: committed (causal) DRMS [m].

Element	Change	Mag 4	Mag 5
reference ($\sigma_\beta = 100$)		24.2 (42.7)	11.5 (16.0)
Comp. prior	$\sigma_\beta = 10$	24.1 (43.9)	11.5 (16.6)
	$\sigma_\beta = 1000$	23.9 (42.1)	11.2 (15.4)
	per column, 100 nT	23.7 (42.7)	11.6 (16.1)
Comp. walk	$\mathbf{Q}^\beta \times 0.01$	26.3 (56.4)	12.7 (20.0)
	$\mathbf{Q}^\beta \times 100$	24.1 (42.7)	12.4 (25.8)
FOGM	$\sigma_S \times 10$	23.9 (38.8)	11.0 (16.6)
Robust kernel	removed	25.4 (42.9)	11.7 (16.5)
	Huber $\delta = 10$	25.6 (42.8)	11.7 (16.5)
Meas. noise	$\sqrt{R} = 65$ nT	24.6 (42.9)	12.7 (23.7)
Process floor	$10^{-16} \mathbf{I}$	33.1 (74.7)	13.1 (19.3)
Relin.	every variable, every epoch	24.1 (42.6)	11.5 (16.0)
Meas. density	all 10 samples per state	26.5 (44.2)	11.7 (16.0)

the attitude-driven regressor to decorrelate, after which further look-ahead adds little. We take 300 s as the default. Second, the causal column is flat: across a twentyfold change in lag the causal error moves by less than a metre on every case. The lag decides only how long a state is revised before it is frozen, not how good the newest state is, so the accuracy bought by look-ahead is entirely smoothing gain, and a designer who cannot afford latency loses exactly the gap between the two columns and nothing else.

F. Ablation

Table VI varies each element of the configuration on line 1007.06, one at a time. The estimator is insensitive at the metre level to all of them. The compensation prior may move an order of magnitude in either direction, or be replaced by a per-column prior that assigns each regressor column 100 nT of measurement contribution, for changes under a metre. Slowing the compensation walk a hundredfold costs about two metres, consistent with a real installation state that drifts in flight; speeding it up costs little. Removing the Huber kernel or widening its threshold costs one to two metres, so robustness to map error helps but does not carry the result. Inflating the assumed measurement noise fivefold, or relinearizing

every variable at every epoch, changes the estimate by less than half a metre; the latter confirms that iSAM2’s selective relinearization gives up nothing here. The largest single effect is the process-noise floor: raising it from 10^{-16} to 10^{-12} times the nominal position channel, i.e. from $10^{-20}\mathbf{I}$ to $10^{-16}\mathbf{I}$, costs 8.9m on Mag 4, which is why Section C treats the floor as a conditioning device to be kept minimal rather than a tuning knob. Attaching all ten raw measurements to each state instead of one costs 2.3m on Mag 4 and nothing on Mag 5, at four times the computation, supporting the 1 Hz state cadence.

G. Computational cost

All timings are single-core, on a laptop-class x86 processor under WSL2, with the measurement factor evaluated through a Python callback; they are upper bounds on a native implementation. At the reference configuration the incremental update takes 6.0ms to 9.0ms per state across the ten line-magnetometer cases, mean 7.8ms, so the 87min line completes in 42s and the margin to the 1 Hz state cadence exceeds two orders of magnitude. The cost scales with the number of retained states, so halving the lag halves it, and the two expensive variants of the ablation bound the alternatives: relinearizing every variable at every epoch costs 68ms per state and attaching all ten raw measurements 31ms, both still real time at 1 Hz but paying roughly ninefold and fourfold for accuracy changes the ablation shows to be, respectively, below half a metre and a two-metre degradation.

VI. Conclusion

This paper posed cold-start magnetic-anomaly navigation as a single estimation problem: an aircraft with an uncalibrated magnetometer installation must recover its position from a scalar reading in which the map signal and the aircraft’s own field are mixed, with no calibration flight and no satellite aiding. The formulation carries the Tolles–Lawson compensation coefficients as time-varying variables of a factor graph alongside the inertial error state, ties the two chains through the map measurement, and solves the graph with an incremental fixed-lag smoother built on iSAM2, at a 300s lag and a 1 Hz state cadence. The estimator produces two outputs, a causal estimate with no look-ahead and a committed estimate revised by the full lag, and we report both throughout, because they answer different questions: the causal output is what a navigator reads in flight, the committed output is what a system that may lag real time delivers.

On the SGL 2020 flight data the answers are as follows. Against a recursive EKF carrying the identical models, the causal output alone is more accurate on six of eight cold-start cases, so windowed relinearization pays before any look-ahead is spent. Against the strongest published causal method, an EKF augmented with an online neural network, the linear-basis smoother is more

accurate on both magnetometers of the representative line with no learned component at all. The committed output is the most accurate estimate on all eight counted cases, between 10.5 and 28.4m where free-inertial drift is hundreds of metres, at 1.5 to 5.2 times the accuracy of the best causal baseline. The lag sweep locates the design knee near 300s, about 20km of track, and shows that the causal output is indifferent to the lag, so the choice of lag is purely a purchase of smoothing accuracy with latency. The ablation finds no delicate setting: one configuration serves every line and both sensors, and the update costs 6.0ms to 9.0ms per state in an unoptimized implementation, two orders of magnitude inside real time.

Several limitations bound the study. Each line is a single physical realization, so every DRMS is a point estimate; the statistical statements are counts across cases, not distributions. The reported covariance of the incremental smoother has not been validated for consistency on flight data, and we do not report uncertainty bounds. The baselines are our own implementations under one shared protocol; the particle-filter reference in particular is reported as unstable under per-case tuning, and we do not claim it represents the best achievable particle filter. None of the lines is a long, straight-and-level operational transit, so performance under the weak regressor excitation of a pure cruise is not established. And the compensation prior that a cold start needs is a measured setting rather than a derived one: we show a broad insensitive range, but give no rule for choosing its scale from installation data.

These limitations set the agenda: consistency evaluation of the reported covariance, a transit-like flight regime, sparser instantiation of the compensation chain in the manner of [49], and a native-factor implementation that would carry the real-time margin from the state cadence down to the raw sensor rate.

Appendix A

Separability of Position and Compensation

The linearized measurement (17) carries a position error only through the map gradient and a compensation error only through the regressor, so over a window the two are distinguishable exactly when no position perturbation and coefficient perturbation produce the same residual sequence. The following condition makes that precise; it supports the discussion of Section A.

Proposition 1:

A position perturbation $\delta\mathbf{p}$ is indistinguishable from a compensation adjustment, hence unobservable from map matching over the window, if and only if $\mathbf{G}\delta\mathbf{p} \in \text{range}(\Psi)$. Consequently the joint state $(\delta\mathbf{p}, \delta\beta)$ is observable if and only if the stacked matrix $[\mathbf{G} \ \Psi]$ has full column rank; equivalently, if and only if all three of the following hold: (i) $\ker \mathbf{G} = \{\mathbf{0}\}$ (the map gradient excites every position direction over the window); (ii) $\ker \Psi = \{\mathbf{0}\}$ (the compensation basis has full column rank); and (iii) $\text{range}(\mathbf{G}) \cap \text{range}(\Psi) = \{\mathbf{0}\}$ (no shared

direction). Condition (iii) alone governs the position–compensation confounding: when it fails, a position shift with $\mathbf{G} \delta \mathbf{p} \neq \mathbf{0}$ is reproduced exactly by a compensation change.

Remark 1:

The condition is geometric: it depends on the ranges of $\mathbf{G}$ and Ψ over the window, hence on the flight path and the local map gradient. It explains the dominant failure mode seen in practice: over a locally flat map $\mathbf{G} \approx \mathbf{0}$ and condition (i) fails, so position is unobservable regardless of the compensation, a map-information floor that bounds every estimator, which we observe on the shortest line in Section V.

Appendix B Pinson Error-State Dynamics

The navigation-error block of Φ_t in (12) follows the Pinson model in a local north-east-down frame [2], [1]. With position, velocity, and attitude errors $(\delta \mathbf{p}, \delta \mathbf{v}, \psi)$ and sensor biases $(\mathbf{b}_a, \mathbf{b}_g)$, the continuous error dynamics are

$$\begin{aligned}\dot{\delta \mathbf{p}} &= \mathbf{F}_{pp} \delta \mathbf{p} + \delta \mathbf{v}, \\ \dot{\delta \mathbf{v}} &= \mathbf{F}_{vp} \delta \mathbf{p} + \mathbf{F}_{vv} \delta \mathbf{v} - [\mathbf{f}^n \times] \psi + \mathbf{C}_b^n \mathbf{b}_a, \\ \dot{\psi} &= \mathbf{F}_{\psi p} \delta \mathbf{p} + \mathbf{F}_{\psi v} \delta \mathbf{v} - [\omega_{in}^n \times] \psi - \mathbf{C}_b^n \mathbf{b}_g,\end{aligned}\quad (25)$$

where $\mathbf{f}^n$ is the specific force, $\mathbf{C}_b^n$ the body-to-nav rotation, ω_{in}^n the inertial rate, $[\cdot \times]$ the skew operator, and the $\mathbf{F}_{\bullet\bullet}$ blocks collect the transport-rate, gravity-gradient, and Coriolis terms. The biases follow first-order Gauss–Markov or random-walk models, as does the scalar magnetic disturbance state S with correlation time τ . The discrete $\Phi_t = \exp(\mathbf{F}_t \Delta t)$ and $\mathbf{Q}_t$ are obtained by standard van Loan discretization [47]. Only $\mathbf{g}_t^\top$, the position row of the linearized measurement (17), couples this block to the map.

Appendix C Tolles–Lawson Basis

The basis row $\mathbf{A}_t$ of (14) is built from the fluxgate direction cosines $\hat{\mathbf{b}} = [u \ v \ w]^\top$ (with $u^2 + v^2 + w^2 = 1$) and the scalar field magnitude B [11], [12]. The permanent moment contributes the three cosines (u, v, w) ; the induced magnetization, symmetric in the field, contributes the six products $B(u^2, uv, uw, v^2, vw, w^2)$; and the eddy-current field, proportional to the rate of change of the cosines, contributes the nine products $B(u\dot{u}, u\dot{v}, \dots, w\dot{w})$. Stacking these gives an eighteen-term basis, optionally augmented by a constant bias, and $c_t = \mathbf{A}_t^\top \beta_t$ is linear in the coefficients β_t , the property that lets them enter the factor graph as ordinary linear-measurement variables. In the cold-start regime the calibration flight that classically fixes β is unavailable, which is precisely why we carry it as a graph variable.

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Appendix D Proof of Proposition 1

Two joint states $(\delta \mathbf{p}, \delta \beta)$ and $(\delta \mathbf{p}', \delta \beta')$ produce identical noise-free predictions iff $\mathbf{G}(\delta \mathbf{p} - \delta \mathbf{p}') = \Psi(\delta \beta' - \delta \beta)$, i.e. iff $[\mathbf{G} \ \Psi][(\delta \mathbf{p} - \delta \mathbf{p}')^\top (\delta \beta - \delta \beta')^\top]^\top = \mathbf{0}$. The unobservable subspace is therefore exactly $\ker[\mathbf{G} \ \Psi]$, and the joint state is observable iff $[\mathbf{G} \ \Psi]$ has full column rank; the first statement of the proposition follows by taking $\delta \beta'$ free and $\delta \mathbf{p}' = \mathbf{0}$.

For the three-condition equivalence: if (i) fails, take $\delta \mathbf{p} \in \ker \mathbf{G} \setminus \{\mathbf{0}\}$, giving the kernel element $(\delta \mathbf{p}, \mathbf{0})$; if (ii) fails, take $\delta \beta \in \ker \Psi \setminus \{\mathbf{0}\}$, giving $(\mathbf{0}, \delta \beta)$; if (iii) fails, take $\mathbf{G} \delta \mathbf{p} = \Psi \delta \beta \neq \mathbf{0}$, giving $(\delta \mathbf{p}, -\delta \beta)$. Conversely, let $(\delta \mathbf{p}, \delta \beta) \neq (\mathbf{0}, \mathbf{0})$ lie in the kernel, so $\mathbf{G} \delta \mathbf{p} = -\Psi \delta \beta$. If $\mathbf{G} \delta \mathbf{p} = \mathbf{0}$ then either $\delta \mathbf{p} \neq \mathbf{0}$ violates (i) or $\delta \beta \neq \mathbf{0}$ violates (ii); if $\mathbf{G} \delta \mathbf{p} \neq \mathbf{0}$ it is a nonzero shared direction, violating (iii). ■

REFERENCES

- [1] P. D. Groves
Principles of GNSS, Inertial, and Multisensor Integrated Navigation Systems, 2nd ed. Boston, MA: Artech House, 2013.
- [2] D. H. Titterton and J. L. Weston

- Strapdown Inertial Navigation Technology*, 2nd ed. Institution of Engineering and Technology, 2004.
- [3] A. Canciani and J. Raquet
Absolute positioning using the Earth's magnetic anomaly field
NAVIGATION: Journal of the Institute of Navigation, vol. 63, no. 2, pp. 111–126, 2016.
 - [4] A. Canciani and J. Raquet
Airborne magnetic anomaly navigation
IEEE Transactions on Aerospace and Electronic Systems, vol. 53, no. 1, pp. 67–80, 2017.
 - [5] M. Muradoğlu, M. T. Johnsson, N. M. Wilson *et al.*
Quantum-assured magnetic navigation achieves positioning accuracy better than a strategic-grade INS in airborne and ground-based field trials
arXiv preprint arXiv:2504.08167, 2025.
 - [6] J. P. Golden
Terrain contour matching (TERCOM): A cruise missile guidance aid
In *Image Processing for Missile Guidance, Proc. SPIE*, vol. 238, 1980, pp. 10–18.
 - [7] L. Hostetler and R. Andreas
Nonlinear Kalman filtering techniques for terrain-aided navigation
IEEE Transactions on Automatic Control, vol. 28, no. 3, pp. 315–323, 1983.
 - [8] P.-J. Nordlund and F. Gustafsson
Marginalized particle filter for accurate and reliable terrain-aided navigation
IEEE Transactions on Aerospace and Electronic Systems, vol. 45, no. 4, pp. 1385–1399, 2009.
 - [9] A. J. Canciani
Magnetic navigation on an F-16 aircraft using online calibration
IEEE Transactions on Aerospace and Electronic Systems, vol. 58, no. 1, pp. 420–434, 2022.
 - [10] R. Angappan, K. Moore, and S. Thoram
Towards unified geophysical data requirements for magnetic navigation (MagNav)
arXiv preprint arXiv:2606.14623, 2026.
 - [11] W. E. Tolles and J. D. Lawson
Magnetic compensation of MH-1A aircraft
Airborne Instruments Laboratory, Tech. Rep. Report 1, 1950.
 - [12] P. Leliak
Identification and evaluation of magnetic-field sources of magnetic airborne detector equipped aircraft
IRE Transactions on Aerospace and Navigational Electronics, vol. ANE-8, no. 3, pp. 95–105, 1961.
 - [13] S. H. Bickel
Small signal compensation of magnetic fields resulting from aircraft maneuvers
IEEE Transactions on Aerospace and Electronic Systems, vol. AES-15, no. 4, pp. 518–525, 1979.
 - [14] A. R. Gnadt *et al.*
Signal enhancement for magnetic navigation challenge problem
arXiv preprint, 2020.
 - [15] A. R. Gnadt
Machine learning-enhanced magnetic calibration for airborne magnetic anomaly navigation
Ph.D. dissertation, Massachusetts Institute of Technology, Cambridge, MA, 2022.
 - [16] P. M. Williams
Aeromagnetic compensation using neural networks
Neural Computing & Applications, vol. 1, no. 3, pp. 207–214, 1993.
 - [17] F. Nerrise, A. S. Sosanya, and P. Neary
Physics-informed calibration of aeromagnetic compensation in magnetic navigation systems using liquid time-constant networks
arXiv preprint, 2024, neurIPS 2023 Machine Learning and the Physical Sciences Workshop.
 - [18] G. Noriega
Aeromagnetic compensation in gradiometry—performance, model stability, and robustness
IEEE Geoscience and Remote Sensing Letters, vol. 12, no. 1, pp. 117–121, 2015.
 - [19] A. Hager, S. Nebendahl, A. Klushyn, J. Krauser, T. H. Bryne, and T. A. Johansen
Airborne magnetic anomaly navigation with neural-network-augmented online calibration
arXiv preprint arXiv:2603.08265, 2026.
 - [20] R. E. Kalman
A new approach to linear filtering and prediction problems
Journal of Basic Engineering, vol. 82, no. 1, pp. 35–45, 1960.
 - [21] N. Bergman
Recursive Bayesian estimation: Navigation and tracking applications
Ph.D. dissertation, Linköping University, Linköping, Sweden, 1999.
 - [22] F. Gustafsson *et al.*
Particle filters for positioning, navigation, and tracking
IEEE Transactions on Signal Processing, vol. 50, no. 2, pp. 425–437, 2002.
 - [23] K. B. Anonsen and O. Hallingstad
Terrain aided underwater navigation using point mass and particle filters
In *IEEE/ION Position, Location, and Navigation Symposium (PLANS)*, 2006, pp. 1027–1035.
 - [24] S. J. Julier and J. K. Uhlmann
Unscented filtering and nonlinear estimation
Proceedings of the IEEE, vol. 92, no. 3, pp. 401–422, 2004.
 - [25] A. Barrau and S. Bonnabel
The invariant extended Kalman filter as a stable observer
IEEE Transactions on Automatic Control, vol. 62, no. 4, pp. 1797–1812, 2017.
 - [26] Y. Bar-Shalom, X.-R. Li, and T. Kirubarajan
Estimation with Applications to Tracking and Navigation. John Wiley & Sons, 2001.
 - [27] F. R. Kschischang, B. J. Frey, and H.-A. Loeliger
Factor graphs and the sum-product algorithm
IEEE Transactions on Information Theory, vol. 47, no. 2, pp. 498–519, 2001.
 - [28] F. Dellaert and M. Kaess
Factor graphs for robot perception
Foundations and Trends in Robotics, vol. 6, no. 1-2, pp. 1–139, 2017.
 - [29] F. Dellaert
Factor graphs and GTSAM: A hands-on introduction
Georgia Institute of Technology, Tech. Rep. GT-RIM-CP&R-2012-002, 2012.
 - [30] B. Triggs, P. F. McLauchlan, R. I. Hartley, and A. W. Fitzgibbon
Bundle adjustment—a modern synthesis
In *Vision Algorithms: Theory and Practice*, 2000, pp. 298–372.
 - [31] G. Grisetti, R. Kümmerle, C. Stachniss, and W. Burgard
A tutorial on graph-based SLAM
IEEE Intelligent Transportation Systems Magazine, vol. 2, no. 4, pp. 31–43, 2010.
 - [32] R. Kümmerle, G. Grisetti, H. Strasdat, K. Konolige, and W. Burgard
g2o: A general framework for graph optimization
In *IEEE International Conference on Robotics and Automation (ICRA)*, 2011, pp. 3607–3613.
 - [33] F. Dellaert and M. Kaess
Square root SAM: Simultaneous localization and mapping via square root information smoothing
The International Journal of Robotics Research, vol. 25, no. 12, pp. 1181–1203, 2006.
 - [34] M. Kaess, A. Ranganathan, and F. Dellaert
iSAM: Incremental smoothing and mapping

- IEEE Transactions on Robotics*, vol. 24, no. 6, pp. 1365–1378, 2008.
- [35] M. Kaess, H. Johannsson, R. Roberts, V. Ila, J. J. Leonard, and F. Dellaert
iSAM2: Incremental smoothing and mapping using the Bayes tree
The International Journal of Robotics Research, vol. 31, no. 2, pp. 216–235, 2012.
- [36] R. M. Eustice, H. Singh, and J. J. Leonard
Exactly sparse delayed-state filters for view-based SLAM
IEEE Transactions on Robotics, vol. 22, no. 6, pp. 1100–1114, 2006.
- [37] G. Sibley, L. Matthies, and G. Sukhatme
Sliding window filter with application to planetary landing
Journal of Field Robotics, vol. 27, no. 5, pp. 587–608, 2010.
- [38] S. Leutenegger, S. Lynen, M. Bosse, R. Siegwart, and P. Furgale
Keyframe-based visual-inertial odometry using nonlinear optimization
The International Journal of Robotics Research, vol. 34, no. 3, pp. 314–334, 2015.
- [39] C. Forster, L. Carlone, F. Dellaert, and D. Scaramuzza
On-manifold preintegration for real-time visual-inertial odometry
IEEE Transactions on Robotics, vol. 33, no. 1, pp. 1–21, 2017.
- [40] V. Indelman, S. Williams, M. Kaess, and F. Dellaert
Information fusion in navigation systems via factor graph based incremental smoothing
Robotics and Autonomous Systems, vol. 61, no. 8, pp. 721–738, 2013.
- [41] T. Qin, P. Li, and S. Shen
VINS-Mono: A robust and versatile monocular visual-inertial state estimator
IEEE Transactions on Robotics, vol. 34, no. 4, pp. 1004–1020, 2018.
- [42] F. W. Lathrop, C. N. Taylor, and A. P. Nielsen
Magnetic sensor compensation utilizing factor graph estimation
IEEE Sensors Journal, vol. 24, no. 15, pp. 23 711–23 722, 2024.
- [43] H. E. Rauch, F. Tung, and C. T. Striebel
Maximum likelihood estimates of linear dynamic systems
AIAA Journal, vol. 3, no. 8, pp. 1445–1450, 1965.
- [44] J. Hollowell
Heli/SITAN: A terrain referenced navigation algorithm for helicopters
Sandia National Laboratories, Albuquerque, NM, USA, Tech. Rep., 1990.
- [45] M. Cowie, N. Wilkinson, and R. Powlesland
Latest development of the TERPROM digital terrain system
In *Proc. IEEE/ION Position, Location and Navigation Symposium*, 2008, pp. 1219–1229.
- [46] H.-A. Loeliger
An introduction to factor graphs
IEEE Signal Processing Magazine, vol. 21, no. 1, pp. 28–41, 2004.
- [47] J. A. Farrell
Aided Navigation: GPS with High Rate Sensors. McGraw-Hill, 2008.
- [48] J. Park and H. Bang
Tight integration of inertial and terrain-referenced navigation under biased odometry and measurements via factor graph optimization
In *Proc. 23rd International Conference on Control, Automation and Systems (ICCAS)*, 2023, pp. 178–183.
- [49] J. Park and H. Bang
Sparse instantiation of bias nodes for factor graph-based terrain-referenced navigation
International Journal of Control, Automation, and Systems, vol. 22, no. 11, pp. 3364–3376, 2024.
- [50] P. J. Huber
Robust estimation of a location parameter
The Annals of Mathematical Statistics, vol. 35, no. 1, pp. 73–101, 1964.
- [51] N. Sünderhauf and P. Protzel
Switchable constraints for robust pose graph SLAM
In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2012, pp. 1879–1884.
- [52] P. Agarwal, G. D. Tipaldi, L. Spinello, C. Stachniss, and W. Burgard
Robust map optimization using dynamic covariance scaling
In *IEEE International Conference on Robotics and Automation (ICRA)*, 2013, pp. 62–69.